

scRNA-seq Report

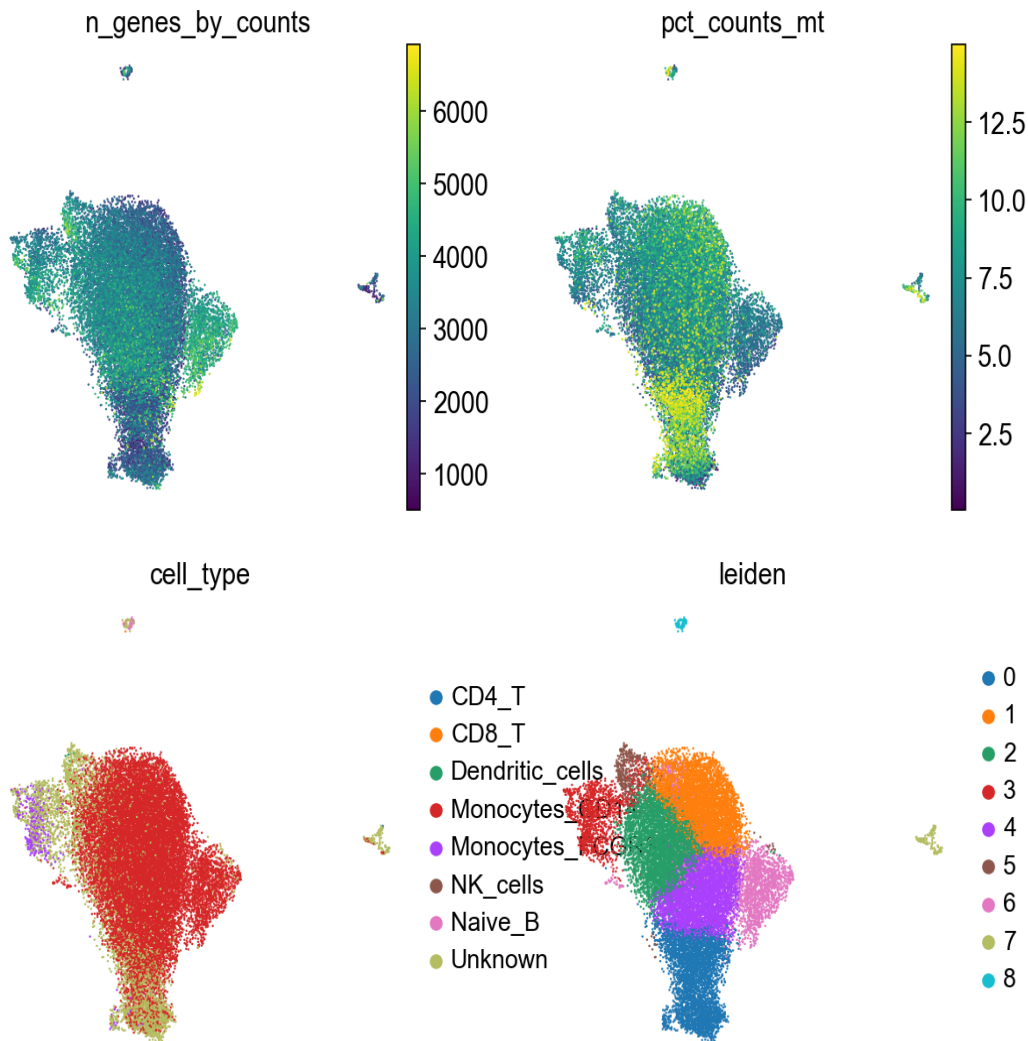
260430-melanoma-monocytes

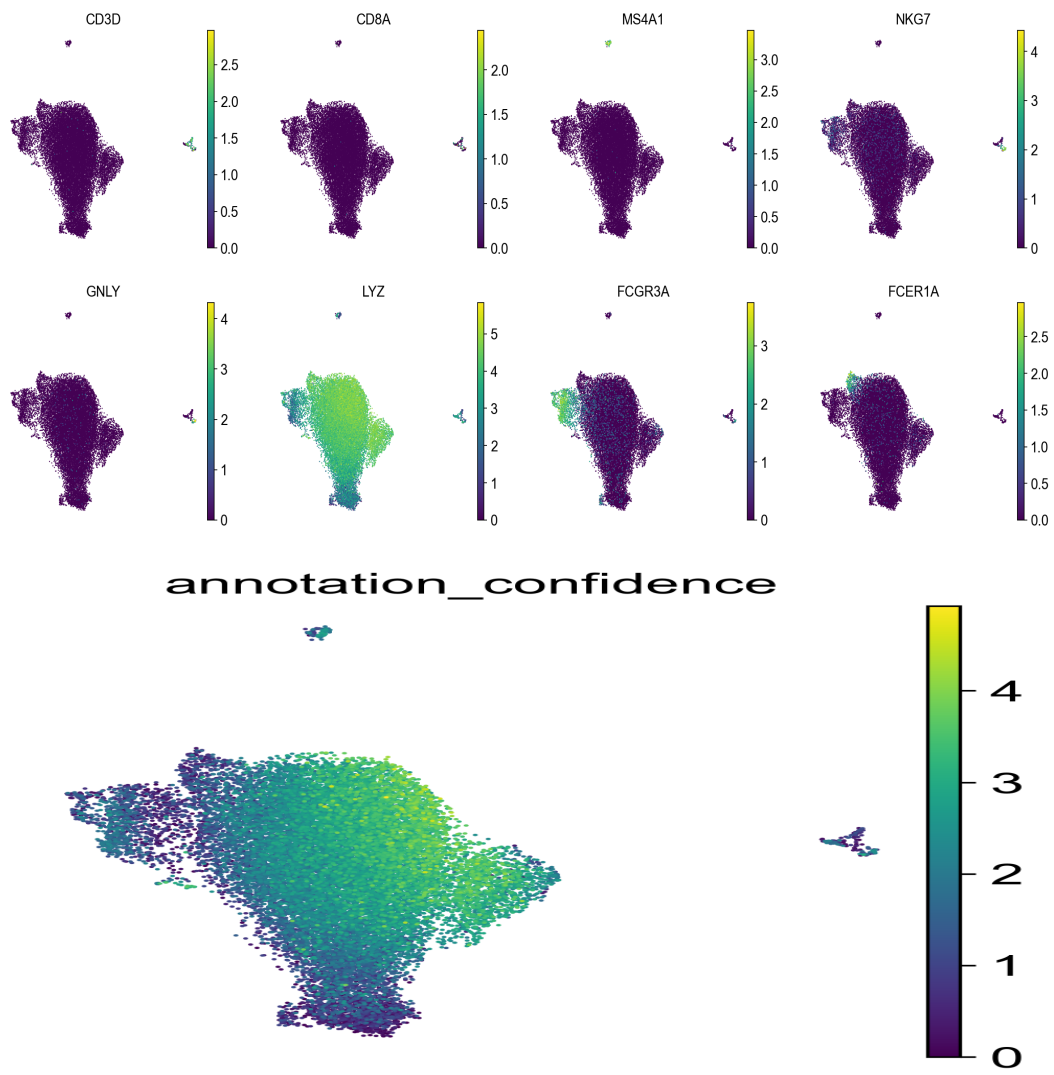
Project: 260430-melanoma-monocytes
Date: 2026-05-02
Cells: 22762
Genes: 2000

1. Quality Control

min_genes_per_cell: 500
max_genes_per_cell: 7000
max_pct_mt: 15.0
min_cells_per_gene: 3

2. UMAP & Clustering





3. Top Marker Genes

cluster	gene	logFC	p_adj
0	MALAT1	1.30	0.0e+00
0	DPYD	0.75	0.0e+00
0	AOAH	0.69	0.0e+00
1	RPS24	1.25	0.0e+00
1	RPLP1	1.20	0.0e+00
1	S100A9	1.67	0.0e+00
2	PID1	1.51	0.0e+00
2	HLA-DRA	1.04	0.0e+00
2	HLA-DPB1	1.24	0.0e+00
3	RPS19	2.32	0.0e+00
3	LST1	2.10	0.0e+00
3	AIF1	2.18	0.0e+00
4	VCAN	0.97	0.0e+00

cluster	gene	logFC	p_adj
4	MEGF9	0.92	0.0e+00
4	DPYD	0.50	0.0e+00
5	CD74	2.13	5.9e-203
5	HLA-DRB1	2.73	5.7e-199
5	HLA-DPB1	3.15	2.1e-194
6	UTY	6.08	0.0e+00
6	NAMPT	2.54	0.0e+00
6	RPS4Y1	6.10	0.0e+00
7	SKAP1	9.45	3.5e-78
7	CD247	9.73	1.0e-66
7	PRKCH	4.62	3.3e-64
8	AFF3	6.31	6.0e-49
8	RALGPS2	5.96	1.1e-46
8	BANK1	8.26	1.5e-46

4. Cluster Annotation Consistency

cluster	cell type	n_cells	fraction
0	Monocytes_CD14	4163	0.53
1	Monocytes_CD14	5485	0.96
2	Monocytes_CD14	4413	0.83
3	Unknown	1321	0.63
4	Monocytes_CD14	4794	0.95
5	Unknown	405	0.81
6	Monocytes_CD14	1949	0.96
7	Unknown	151	0.88
8	Naive_B	81	0.54

5. Differential Expression & GO Analysis

Differential expression (DEG) analysis identifies genes that are significantly up- or down-regulated between clusters or biological conditions (e.g. healthy vs disease). These DEGs are subsequently used for Gene Ontology (GO) enrichment analysis to identify biological processes, molecular functions, and cellular components associated with each cluster or condition. Typical workflow: 1. Identify marker genes (cluster vs rest) 2. Perform DEG (condition vs condition) 3. Run GO enrichment on significant genes Thus, DEG provides the gene list, and GO provides biological interpretation.

6. Parameters

QC - min_genes_per_cell: 500
QC - max_genes_per_cell: 7000
QC - max_pct_mt: 15.0

QC - min_cells_per_gene: 3
Analysis - n_top_hvgs: 2000
Analysis - n_pcs: 40
Analysis - neighbor_k: 15
Analysis - leiden_resolution: 0.6